

Universal Relation Followed by a Function

1 Definition

Definition 1 (De Morgan formulas). A *De Morgan formula* is a rooted tree whose internal vertices are fan-in-two $\wedge$ - or $\vee$ -gates and whose leaves are literals x_i or $\neg x_i$. For a Boolean function f , let $L(f)$ be the minimum number of leaves of such a formula computing f , and let $D(f)$ be its minimum depth. Since every depth- d formula has at most 2^d leaves,

$$D(f) \geq \log L(f). \quad (1)$$

Definition 2 (Block composition). Let $f : \{0, 1\}^m \rightarrow \{0, 1\}$ and $g : \{0, 1\}^n \rightarrow \{0, 1\}$ be nonconstant. For an $m \times n$ Boolean matrix $X = (X_1, \dots, X_m)$, define

$$(f \diamond g)(X) = f(g(X_1), \dots, g(X_m)). \quad (2)$$

Substituting a formula for g at every input leaf of a formula for f gives

$$D(f \diamond g) \leq D(f) + D(g), \quad L(f \diamond g) \leq L(f)L(g). \quad (3)$$

Definition 3 (Karchmer–Wigderson relation). For a nonconstant $f : \{0, 1\}^n \rightarrow \{0, 1\}$, the relation KW_f is

$$KW_f = \{(x, y, i) : x \in f^{-1}(1), y \in f^{-1}(0), x_i \neq y_i\}. \quad (4)$$

Alice receives x , Bob receives y , and they must output such an index i . Writing $CC(R)$ for deterministic communication complexity, the Karchmer–Wigderson correspondence states

$$D(f) = CC(KW_f), \quad L(f) = \min_{\Pi \text{ solves } KW_f} \text{leaves}(\Pi). \quad (5)$$

Moreover, $KW_{f \diamond g}$ is denoted by $KW_f \diamond KW_g$.

Definition 4 (Universal relation). For $n \geq 1$, the *universal relation* U_n is the search problem in which Alice receives $x \in \{0, 1\}^n$, Bob receives $y \in \{0, 1\}^n$, and they are promised that $x \neq y$. Their task is to output an index $j \in [n]$ such that $x_j \neq y_j$. Replacing one factor by U gives the relaxations $KW_f \diamond U_n$ and $U_m \diamond KW_g$.

Definition 5 (Strong composition). Let $X, Y \in (\{0, 1\}^n)^m$ be valid inputs for $KW_{f \diamond g}$, and set $a_i = g(X_i)$ and $b_i = g(Y_i)$. In the ordinary relation $KW_f \diamond KW_g = KW_{f \diamond g}$, the players may output any pair (i, j) with $X_{i,j} \neq Y_{i,j}$. In the *strong composition* $KW_f \circledast KW_g$, they must output (i, j) satisfying

$$a_i \neq b_i \quad \text{and} \quad X_{i,j} \neq Y_{i,j}. \quad (6)$$

Hence the output row must itself be a valid instance of KW_g . Strong composition therefore enforces the two-stage direct-sum strategy: first identify an outer solution i , then solve the inner relation in row i .

Conjecture 1 (Karchmer–Raz–Wigderson). *There is a universal constant $C > 0$ such that, for all nonconstant $f : \{0, 1\}^m \rightarrow \{0, 1\}$ and $g : \{0, 1\}^n \rightarrow \{0, 1\}$,*

$$D(f \diamond g) \geq D(f) + D(g) - C \log(mn). \quad (7)$$

Equivalently,

$$CC(KW_f \diamond KW_g) \geq CC(KW_f) + CC(KW_g) - C \log(mn). \quad (8)$$

Together with (3), this asserts additive depth under composition up to a logarithmic loss.

2 Current best result

When $m = \omega(\log^2 n)$ and $n = \omega(\sqrt{m})$, a random nonconstant $g : \{0, 1\}^n \rightarrow \{0, 1\}$ satisfies, with probability $1 - o(1)$,

$$CC(U_m \diamond KW_g) \geq m + n - O(\sqrt{m}). \quad (9)$$

Thus a near-additive bound is known for almost all inner functions in this parameter range, but not for every g .

3 Problem formalization

For every nonconstant $g : \{0, 1\}^n \rightarrow \{0, 1\}$, and for the relevant ranges of m, n , prove

$$CC(U_m \diamond KW_g) \geq m + CC(KW_g) - o(\min\{m, CC(KW_g)\}). \quad (10)$$